



Indian Institute of Information Technology, Surat  
Department of Electronics & Communication Engineering  
**Mid Semester Examination, 4<sup>th</sup> Sem, 2024-25**  
Subject Name : Analog & Digital Communication (EC-407)

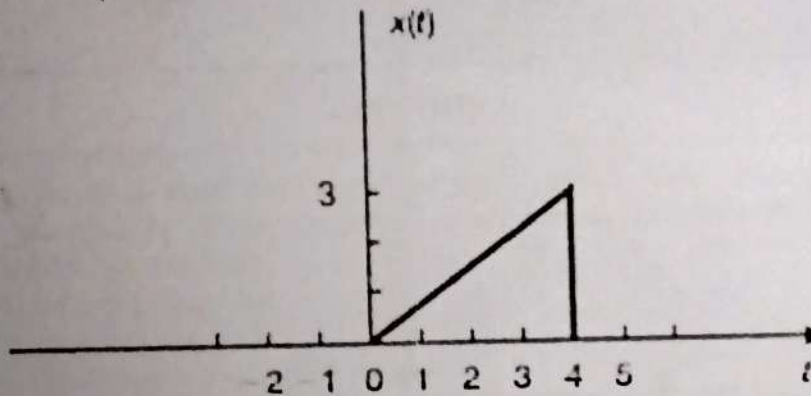
Timing: 10:00 AM to 11:30 AM

Date: 10/03/2025

Max Marks: 30

- Note: 1. All questions are compulsory.  
2. Assume suitable data wherever necessary.  
3. Draw suitable diagrams when required.

1. (a) A continuous-time signal  $x(t)$  is shown in Figure. Sketch and label each of the following signals. (a)  $x(t-2)$ ; (b)  $x(2t)$ ; (c)  $x(t/2)$ ; (d)  $x(-t)$  (4)



- (b) Find the Fourier transform for the given  $x(t)$  which is defined by: (4)

$$x(t) = \begin{cases} 1 & |t| < a \\ 0 & |t| > a \end{cases}$$

2. (a) State and verify the time convolution theorem for the Fourier transform (4)

- (b) For the four point sequence  $x(n) = \{3, 2, 1, 0\}$ , Compute the Discrete Fourier transform (DFT) based on the output find the IDFT for the same. (8)

3. (a) Derive the expression of total power in AM. (3)

- (b) Explain balanced modulator circuit in detail with block diagram. (3)

- (c) A modulating signal  $20 \cos(20\pi \cdot 10^3 t)$  is used to modulate a carrier signal  $40 \cos(40\pi \cdot 10^5 t)$ . Find the modulating index, amplitude of each sideband, frequencies of sideband components, bandwidth, efficiency and equation of AM wave. Draw single- and double-sided spectrum of the AM wave. (4)